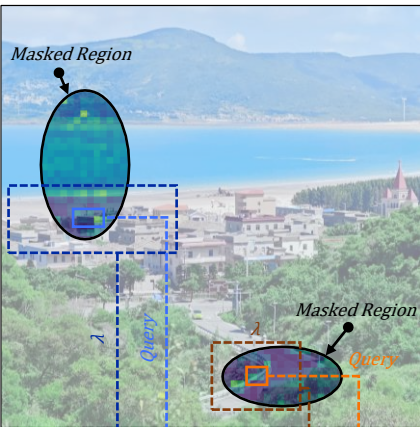


# Latent Representation



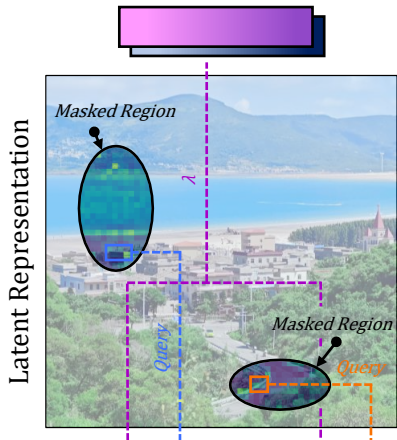
$$\begin{bmatrix} \text{blue} & \text{blue} \\ \text{blue} & \text{blue} \end{bmatrix} = \begin{bmatrix} \text{blue} & \text{blue} & \text{blue} \\ \text{blue} & \text{blue} & \text{blue} \end{bmatrix} * \begin{bmatrix} \text{blue} \\ \text{blue} \end{bmatrix}$$

$$\begin{bmatrix} \text{orange} & \text{orange} \\ \text{orange} & \text{orange} \end{bmatrix} = \begin{bmatrix} \text{orange} & \text{orange} & \text{orange} \\ \text{orange} & \text{orange} & \text{orange} \end{bmatrix} * \begin{bmatrix} \text{orange} \\ \text{orange} \end{bmatrix}$$

Interacted Features

(a) Local- $\lambda$

# LCG Embedding



$$\begin{bmatrix} \text{blue} & \text{blue} \\ \text{blue} & \text{blue} \end{bmatrix} = \begin{bmatrix} \text{purple} & \text{purple} & \text{purple} \\ \text{purple} & \text{purple} & \text{purple} \end{bmatrix} * \begin{bmatrix} \text{blue} \\ \text{blue} \end{bmatrix}$$

$$\begin{bmatrix} \text{orange} & \text{orange} \\ \text{orange} & \text{orange} \end{bmatrix} = \begin{bmatrix} \text{purple} & \text{purple} & \text{purple} \\ \text{purple} & \text{purple} & \text{purple} \end{bmatrix} * \begin{bmatrix} \text{orange} \\ \text{orange} \end{bmatrix}$$

Interacted Features

(b) Interactive- $\lambda$